Trigonometry

1. Trigonometry

1.1. Domain/Range of trig functions

Identities	Domain	Range
$\sin\theta$	$\mathbb{R}$	$[-1, 1]$
$\cos\theta$	$\mathbb{R}$	$[-1, 1]$
$\tan\theta$	$\mathbb{R} - \left\{(2n+1)\frac{\pi}{2}\right\}$	$\mathbb{R}$
$\cot\theta$	$\mathbb{R} - \{n\pi\}$	$\mathbb{R}$
$\csc\theta$	$\mathbb{R} - \{n\pi\}$	$\mathbb{R} - (-1, 1)$
$\sec\theta$	$\mathbb{R} - \left\{(2n+1)\frac{\pi}{2}\right\}$	$\mathbb{R} - (-1, 1)$

1.2. Trigonometric Formulas

1. $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$
2. $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$
3. $\tan(A \pm B) = \dfrac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$
4. $\cot(A \pm B) = \dfrac{\cot A \cot B \mp 1}{\cot B \pm \cot A}$
5. $\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$
6. $\sin(A + B) - \sin(A - B) = 2 \cos A \sin B$
7. $\cos(A + B) + \cos(A - B) = 2 \cos A \cos B$
8. $\cos(A + B) - \cos(A - B) = -2 \sin A \sin B$
9. $\sin(A + B) \cdot \sin(A - B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$

10. $\cos(A + B) \cdot \cos(A - B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$

11. $\sin C + \sin D = 2\sin\left(\frac{C + D}{2}\right) \cdot \cos\left(\frac{C - D}{2}\right)$

12. $\sin C - \sin D = 2\cos\left(\frac{C + D}{2}\right) \cdot \sin\left(\frac{C - D}{2}\right)$

13. $\cos C + \cos D = 2\cos\left(\frac{C + D}{2}\right) \cdot \cos\left(\frac{C - D}{2}\right)$

14. $\cos C - \cos D = 2\sin\left(\frac{C + D}{2}\right) \cdot \sin\left(\frac{D - C}{2}\right)$

15. $\sin 2A = 2\sin A\cos A = \frac{2\tan A}{1 + \tan^2 A}$

16. $\cos 2A = \cos^2 A - \sin^2 A = 1 - 2\sin^2 A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$

17. $\tan 2A = \frac{2\tan A}{1 - \tan^2 A}$

18. $\cot 2A = \frac{\cot^2 A - 1}{2\cot A}$

19. $\sin 3A = 3\sin A - 4\sin^3 A$

20. $\cos 3A = 4\cos^3 A - 3\cos A$

21. $\tan 3A = \frac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$

22. $\cot 3A = \frac{\cot^3 A - 3\cot A}{3\cot^2 A - 1}$

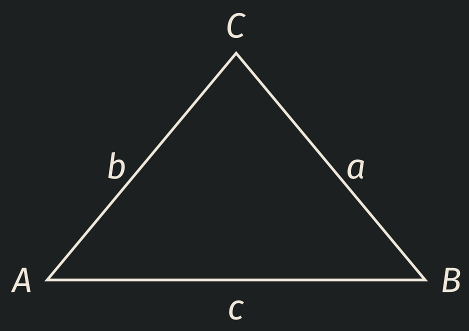
23. $\tan^2 A = \frac{1 - \cos 2A}{1 + \cos 2A}$

1.3. Value of θ

1. $\sin\theta = \sin\alpha \implies \theta = n\pi + (-1)^n\alpha$
2. $\cos\theta = \cos\alpha \implies \theta = 2n\pi \pm \alpha$
3. $\tan\theta = \tan\alpha \implies \theta = n\pi + \alpha$
4. $\sin\theta = 0 \implies \theta = n\pi$
5. $\cos\theta = 0 \implies \theta = n\pi + \frac{\pi}{2} = (2n + 1)\frac{\pi}{2}$
6. $\tan\theta = 0 \implies \theta = n\pi$
7. $\sin\theta = 1 \implies \theta = 2n\pi + \frac{\pi}{2} = (4n + 1)\frac{\pi}{2}$

8. $\cos\theta = 1 \implies \theta = 2n\pi$

2. Properties of a triangles



2.1. Area of a triangle

1. $\Delta = \frac{1}{2}ab\sin C = \frac{1}{2}bc\sin A = \frac{1}{2}ca\sin B$
2. $\Delta = \sqrt{s(s-a)(s-b)(s-c)} \quad s = \frac{a+b+c}{2}$

2.2. Law of Sines:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

where circumradius, $R = \frac{abc}{4\Delta}$

2.3. Law of Cosines:

1. $a^2 = b^2 + c^2 - 2bc\cos A \implies \cos A = \frac{b^2 + c^2 - a^2}{2bc}$
2. $b^2 = a^2 + c^2 - 2ac\cos B \implies \cos B = \frac{a^2 + c^2 - b^2}{2ac}$

3. $c^2 = a^2 + b^2 - 2ab\cos C \implies \cos C = \dfrac{a^2 + b^2 - c^2}{2ab}$

2.4. Projection Rule:

1. $a = b\cos C + c\cos B$
2. $b = c\cos A + a\cos C$
3. $c = a\cos B + b\cos A$

2.5. Napier's analogy:

1. $\tan\left(\dfrac{B-C}{2}\right) = \dfrac{b-c}{b+c}\cot\dfrac{A}{2}$
2. $\tan\left(\dfrac{C-A}{2}\right) = \dfrac{c-a}{c+a}\cot\dfrac{B}{2}$
3. $\tan\left(\dfrac{A-B}{2}\right) = \dfrac{a-b}{a+b}\cot\dfrac{C}{2}$

2.6. Half Angle Formulas:

1. $\sin\left(\dfrac{A}{2}\right) = \sqrt{\dfrac{(s-b)(s-c)}{bc}}$
2. $\cos\left(\dfrac{A}{2}\right) = \sqrt{\dfrac{s(s-a)}{bc}}$
3. $\tan\left(\dfrac{A}{2}\right) = \sqrt{\dfrac{(s-b)(s-c)}{s(s-a)}} = \dfrac{r}{(s-a)}$

   where inradius, $r = 4R\sin\left(\dfrac{A}{2}\right)\sin\left(\dfrac{B}{2}\right)\sin\left(\dfrac{C}{2}\right)$

3. Inverse Trigonometry

3.1. Domain/Range of inverse trig functions

Identities	Domain	Range
$\sin^{-1} x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} x$	$\mathbb{R}$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cot^{-1} x$	$\mathbb{R}$	$(0, \pi)$
$\csc^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$
$\sec^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$[0, \pi] - \left\{\frac{\pi}{2}\right\}$

3.2. Inverse Trigonometric Formulas

1. $\sin^{-1} x = \csc^{-1}\left(\frac{1}{x}\right)$
2. $\cos^{-1} x = \sec^{-1}\left(\frac{1}{x}\right)$
3. $\tan^{-1} x = \cot^{-1}\left(\frac{1}{x}\right), x > 0$

   $= \cot^{-1}\left(\frac{1}{x}\right) - \pi, x < 0$

   $= 0, x = 0$
4. $\csc^{-1} x = \sin^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{2} - \sec^{-1} x$
5. $\sec^{-1} x = \cos^{-1}\left(\frac{1}{x}\right) = \frac{\pi}{2} - \csc^{-1} x$
6. $\cot^{-1} x = \frac{\pi}{2} - \tan^{-1}(x)$
7. $\sin^{-1}(-x) = -\sin^{-1} x$
8. $\cos^{-1}(-x) = \pi - \cos^{-1} x$
9. $\tan^{-1}(-x) = -\tan^{-1} x$

10. $\cot^{-1}(-x) = \pi - \cot^{-1} x$
11. $\csc^{-1}(-x) = -\csc^{-1} x$
12. $\sec^{-1}(-x) = \pi - \sec^{-1} x$
13. $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$
14. $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$
15. $\csc^{-1} x + \sec^{-1} x = \frac{\pi}{2}$
16. $\sin^{-1} x \pm \sin^{-1} y = \sin^{-1}\left(x\sqrt{1-y^2} \pm y\sqrt{1-x^2}\right)$
17. $\cos^{-1} x \pm \cos^{-1} y = \cos^{-1}\left(xy \mp \sqrt{1-x^2}\sqrt{1-y^2}\right)$
18. $$\begin{aligned} \tan^{-1} x + \tan^{-1} y &= \tan^{-1}\left(\frac{x+y}{1-xy}\right), xy < 1 \\ &= \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right), xy > 1, x > 0, y > 0 \\ &= -\pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right), xy > 1, x < 0, y < 0 \end{aligned}$$
19. $\tan^{-1} x - \tan^{-1} y = \tan^{-1}\left(\frac{x-y}{1+xy}\right)$
20. $\tan^{-1} x \pm \tan^{-1} y \pm \tan^{-1} z = \tan^{-1}\left(\frac{x \pm y \pm z \mp xyz}{1 \mp xy \mp yz \mp zx}\right)$
21. $2\sin^{-1} x = \cos^{-1}\left(1-2x^2\right) = \sin^{-1}\left(2x\sqrt{1-x^2}\right)$
22. $2\cos^{-1} x = \cos^{-1}\left(2x^2-1\right)$
23. $2\tan^{-1} x = \tan^{-1}\left(\frac{2x}{1-x^2}\right) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$
24. $3\sin^{-1} x = \sin^{-1}\left(3x-4x^3\right)$
25. $3\cos^{-1} x = \cos^{-1}\left(4x^3-3x\right)$